

Danush Arun

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Summary

Systems-first AI engineer who designs, builds, and ships production systems end-to-end — fast. Specialises in agentic AI architecture and real-time speech intelligence. Equally fluent in original ML research and Kubernetes deployment. Complete ownership: architecture, implementation, testing, and outcome — no hand-offs, no half-built systems.

Professional Experience

DriveX Mobility Private Limited Feb 2025 – Present

Technical Associate Program Manager / Software Engineer Bengaluru, India

MIRA — Multilingual Intelligent Real-time Agent Oct 2025 – Feb 2026

- Designed the architecture of a production outbound voice AI agent serving live leads in 5 Indian languages at sub-100 ms latency; reduced first-response time from 7 s to under 500 ms across 4 architectural iterations over 68 days.
- Owned the full system — real-time speech pipeline, distributed call orchestration, post-call intent classification, CRM synchronisation, and agentic WhatsApp coordination with auto-group creation and a live bot.

AIDEN — AI-Driven ENgagement Analytics Mar 2026

- Shipped a full-stack call analytics platform in 31 days that gave sales management their first data-driven view into agent SOP adherence, conversation quality, and emotional engagement across every call.
- Built speaker-diarized transcription, paralinguistic emotion scoring, and parallel LLM analysis across 8 SOP dimensions; surfaced as a React dashboard embedded in Zoho CRM; deployed to Kubernetes with E2E coverage.

Vanguard, Veronica, FuryX & Quality Engineering

- Built Vanguard, an autonomous web testing agent pairing a local VLM with Playwright to continuously validate the DriveX corporate website end-to-end without scripted test cases.
- Shipped Veronica (AI vehicle inspection, 1,000+ parts) and FuryX (Chrome Extension + Strapi CMS portal); owned QE — JMeter load testing and Selenium ERP automation.

Formula Manipal May 2022 – Jan 2024

Operations Lead Manipal, India

- Led autonomous path planning for the FM23e EV; track-side testing improved control accuracy by 40%; 1st place in Cost & Manufacturing at Formula Bharat 2024; negotiated INR 60,00,000 in sponsorships.

Tescom — Hardware Engineering Internship (2024): Engineered a custom PCB testing solution and optimised assembly line logic to increase error detection accuracy and reduce operational downtime.

Research & Projects

Wave Field Attention — Sub-quadratic Attention Mechanism Research Mar 2026

- Derived an $O(n \log n)$ attention mechanism replacing standard $O(n^2)$ softmax — $\sim 200,000\times$ theoretical speedup at 1 M tokens; grounded in the Alman-Yu lower bound (ICLR 2025). Produced a complete research paper.

Computational Physics Engine & EMI Shield Designer Dec 2024 – Present

- Vectorised Schelkunoff EM shielding simulation for multi-phase composites; React Native UI for frequency sweeps (100 kHz–10 GHz) and 20-year service life prediction; Jenkins CI/CD, SonarQube quality gates.

Technical Skills

AI & ML: Agentic orchestration, real-time voice pipelines, MCP, tool use & structured outputs, speaker diarization, streaming ASR, RAG, computer vision (YOLOv8), VLM-based testing, Pipecat, Qiskit
Engineering: Python (Expert), Django, FastAPI, React, TypeScript, Next.js, SQL, C++, JavaScript, Bash, React Native
Infrastructure & QE: Docker, Kubernetes, Celery, Redis, PostgreSQL, AWS S3, Jenkins, SonarQube, Playwright, Selenium, JMeter
Integrations: Zoho CRM, VoBiz Telephony, WhatsApp Business API, Sarvam AI, Hume AI, Strapi CMS, MetaX

Education & Certifications

Manipal Institute of Technology — B.Tech in Electrical and Electronics Engineering Manipal, India · 2025
IIT Roorkee (iHUB) — Executive PG Certificate in Full Stack Development Roorkee, India · 2025
Certifications: Python for Data Science — IBM • SQL for Data Science — Coursera • Cloud Computing — IBM